

Mathematical Problems for the Next Century¹

U. *I. Arnold, on behalf of the International Mathematical Union, has written to a number of mathematicians with a suggestion that they describe some great problems for the next century. This report is my response.*

Arnold's invitation is inspired in part by Hilbert's list of 1900 (see, e.g., [Browder, 1976]) and I have used that list to help design this essay.

I have listed 18 problems, chosen with these criteria:

1. Simple statement. Also preferably mathematically precise.
2. Personal acquaintance with the problem. I have not found it easy.
3. A belief that the question, its solution, partial results, or even attempts at its solution are likely to have great importance for mathematics and its development in the next century.

Some of these problems are well known. In fact, included are what I believe to be the three greatest open problems of mathematics: the Riemann Hypothesis, the Poincaré Conjecture, and "Does $P = NP$?" Besides the Riemann Hypothesis, one below is on Hilbert's list (Hilbert's 16th Problem). There is a certain overlap with my earlier paper "Dynamics retrospective, great problems, attempts that failed" [Smale, 1991].

Let us begin.

Problem 1: The Riemann Hypothesis

Of the zeros of the Riemann zeta function, defined by analytic continuation from

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \quad \operatorname{Re}(s) > 1,$$

are those which are in the critical strip $0 \leq \operatorname{Re}(s) \leq 1$ all on the line $\operatorname{Re}(s) = \frac{1}{2}$?

This was Problem #8 on Hilbert's list. There are many fine books on the zeta function and the Riemann hypothesis which are easy to locate. I leave the matter at this.

Problem 2: The Poincaré Conjecture

Suppose that a compact connected 3-dimensional manifold has the property that every circle in it can be deformed to a point. Then must it be homeomorphic to the 3-sphere?

The n -sphere is the space

$$S^n = \{x \in \mathbb{R}^{n+1} \mid \|x\| = 1\}, \quad \|x\|^2 = \sum_{i=1}^{n+1} x_i^2.$$

A compact n -dimensional manifold can be thought of as a closed bounded n -dimensional surface (differentiable and non-singular) in some Euclidean space.

¹Lecture given on the occasion of Arnold's 60th birthday at the Fields Institute, Toronto, June 1997.

The n -dimensional Poincaré conjecture asserts that a compact n -dimensional manifold M having the property that every map $f: S^k \rightarrow M$, $k < n$ (or equivalently, $k \leq n/2$) can be deformed to a point, must be homeomorphic to S^n .

Henri Poincaré studied these problems in his pioneering papers in topology. Poincaré in 1900 (see [Poincaré, 1953], pp. 338–370) announced a proof of the general n -dimensional case. Subsequently (in 1904) he found a counterexample to his first version of the statement ([Poincaré, 1953], pp. 435–498). In the second paper he limits himself to $n = 3$, and states the 3-dimensional case as the problem above (not actually as a “conjecture”).

My own relationship with this problem is described in the story [Smale, 1990]. There I wrote,

I first heard of the Poincaré conjecture in 1955 in Ann Arbor at the time I was writing a thesis on a problem of topology. Just a short time later, I felt that I had found a proof (3 dimensions). Hans Samelson was in his office, and very excitedly I sketched my ideas to him. . . . After leaving the office, I realized that my “proof” hadn’t used any hypothesis on the 3-manifold.

In 1960, “on the beaches of Rio,” I gave an affirmative answer to the n -dimensional Poincaré conjecture for $n > 4$. In 1983, Mike Freedman gave an affirmative answer for $n = 4$. (Note: for $n > 4$, I proved the stronger result that M was the smooth union of two balls, $M = D^n \cup D^n$; that result is unproved for $n = 4$, today.)

For background on these matters, besides the above references, see [Smale, 1963].

Many other mathematicians after Poincaré have claimed proofs of the 3-dimensional case. See [Taubes, 1987] for an account of some of these attempts.

A reason that Poincaré’s conjecture is fundamental in the history of mathematics is that it helped give focus to a manifold as an object of study. In this way, Poincaré influenced much of 20th-century mathematics with its attention to geometric objects, including eventually algebraic varieties, Riemannian manifolds, etc.

I hold the conviction that there is a comparable phenomenon today in the notion of a “polynomial-time algorithm.” Algorithms are becoming worthy of analysis in their own right, not merely as a means to solve other problems. Thus I am suggesting that as the study of the set of solutions of an equation (e.g., a manifold) played such an important role in 20th-century mathematics, the study of finding the solutions (e.g., an algorithm) may play an equally important role in the next century.

Problem 3: Does $P = NP$?

I sometimes consider this problem as a gift to mathematics from computer science. It may be useful to put it into a form which looks more like traditional mathematics.

Towards this end, first consider the Hilbert Nullstellensatz over the complex numbers. Thus let $f_1, \dots, f_k$ be complex polynomials in n variables; we are asked to decide if they

have a common zero $\zeta \in \mathbb{C}^n$. The Nullstellensatz asserts that this is not the case if and only if there are complex polynomials $g_1, \dots, g_k$ in n variables satisfying

$$\sum_{i=1}^k g_i f_i = 1 \quad (1)$$

as an identity of polynomials.

The effective Nullstellensatz as established by Brownawell (1987) and others, states that in the above one may assume that the degrees of the g_i satisfy

$$\deg g_i \leq \max(3, D)^n, \quad D = \max \deg f_i.$$

With this degree bound the decidability problem becomes one of linear algebra. Given the coefficients of the f_i , one can check if (1) has a solution in the coefficients of the g_i . Thus one has an algorithm to decide the Nullstellensatz. The number of arithmetic steps required grows exponentially in the number of coefficients of the f_i (the input size).

Conjecture (over $\mathbb{C}$). *There is no polynomial-time algorithm for deciding the Hilbert Nullstellensatz over $\mathbb{C}$.*

A polynomial-time algorithm is one in which the number of arithmetic steps is bounded by a polynomial in the number of coefficients of the f_i .

To make mathematical sense of this conjecture, one has need of a formal definition of algorithm. In this context, the traditional definition of Turing machine makes no sense. In [Blum-Shub-Smale, 1989] a satisfactory definition is proposed, and the associated theory is exposed in [Blum-Cucker-Shub-Smale (or BCSS), 1997].

Very briefly, a machine over $\mathbb{C}$ has as inputs a finite string of complex numbers, and the same for states and outputs. Computations on states include arithmetic operations and shifts on the string. Finally, a branch operation on “ $x_1 = 0$?” is provided.

The size of an input is the number of elements in the input string. The time of a computation is the number of machine operations used in the passage from input to output. Thus, a polynomial-time algorithm over $\mathbb{C}$ is well-defined.

Note that all that has been said about the machines and the conjecture use only the structure of $\mathbb{C}$ as a field, and hence both the machines and the conjecture are meaningful over any field. In particular, if the field is $\mathbb{Z}_2$ of two elements, we have the Turing machines.

Consider the decision problem. Input k polynomials in n variables with coefficients in $\mathbb{Z}_2$. Is there a common zero $\zeta \in (\mathbb{Z}_2)^n$?

Conjecture. *There is no polynomial-time algorithm over $\mathbb{Z}_2$ deciding this problem.*

This is a simple reformulation of the classic conjecture $P \neq NP$.

In the above I have bypassed the basic ideas and theorems related to NP-completeness. For the classic case of

Cook and Karp, see [Garey-Johnson, 1979], and for the theory over an arbitrary field, see BCSS.

Problem 4: Integer Zeroes of a Polynomial

Let me start by defining a diophantine invariant τ motivated by complexity theory. A program for a polynomial $f \in \mathbb{Z}[t]$ of one variable with integer coefficients is the object $(1, t, u_1, \dots, u_k)$, where $u_k = f$, and for all ℓ , $u_\ell = u_i \circ u_j$, $i, j < \ell$, and $\circ$ is $+$ or $-$ or $\times$. Here $u_0 = t$, $u_{-1} = 1$. Then $\tau(f)$ is the minimum of k over all such programs.

Is the number of distinct integer zeroes of f polynomially bounded by $\tau(f)$? In other words, is

$$Z_a(f) \leq \tau(f)^c \quad \text{all } f \in \mathbb{Z}[t],$$

where $Z_a(f)$ is the number of distinct integer zeros of f and c is a universal constant?

Mike Shub and I discovered this problem in our complexity studies. We proved that an affirmative answer implied the intractibility of the nullstellensatz as a decision problem over $\mathbb{C}$ and thus $P \neq NP$ over $\mathbb{C}$. See [Shub-Smale, 1995] and also BCSS.

Since the degree of f is less than or equal to $2^{\tau+1}$, $\tau = \tau(f)$, there are no more than $2^{\tau+1}$ zeros altogether.

For Chebyshev polynomials, the number of distinct real zeros grows exponentially with τ .

Many of the classic diophantine problems are in two or more variables. This problem asks for an estimate in just one variable, and nevertheless seems not so easy.

Here is a related problem. A program for an integer m is the object $(1, m_1, \dots, m_\ell)$, where $m_\ell = m$, $m_0 = 1$, $m_q = m_i \circ m_j$, $i, j < q$, and $\circ = +, -$ or $\times$. Then let $\tau(m)$ be the minimum of ℓ , over all such programs. Thus $\tau(m)$ represents the shortest way to build up an integer m starting from 1 using plus, minus, and times.

Problem: Is there a constant c such that $\tau(k!) \leq (\log k)^c$ for all integers k ? One might expect this to be false, so that $k!$ is "hard to compute"; see [Shub-Smale, 1995].

Problem 5: Height Bounds for Diophantine Curves

Can the feasibility of a diophantine equation $f(x, y) = 0$ (input $f \in \mathbb{Z}[u, v]$) be decided in time 2^{s^c} where c is a universal constant?

Here $s = s(f)$ is the size of f , defined by

$$s(f) = \sum_{|\alpha| \leq d} (\log |a_\alpha| + 1), \quad f(x, y) = \sum_{|\alpha| \leq d} a_\alpha x^{\alpha_1} y^{\alpha_2},$$

$\alpha = (\alpha_1, \alpha_2)$, and $|\alpha| = \alpha_1 + \alpha_2$, $\alpha_i \geq 0$.

Moreover, f is feasible if there exist integers x, y with $f(x, y) = 0$. The Turing model of computation is supposed.

This problem is posed essentially in [Cucker, Koiraan, and Smale, 1997]. The size $s(f)$ is a version of the "height" of f . Conjectured height bounds for example as in [Lang, 1991] could prove helpful in settling this problem. See also [Manders-Adleman, 1978].

Problem 6: Finiteness of the Number of Relative Equilibria in Celestial Mechanics

Given positive real numbers $m_1, \dots, m_n$ as the masses in the n -body problem of celestial mechanics, is the number of relative equilibria finite?

The problem is in Wintner's book (1941) on celestial mechanics. A relative equilibrium is a solution to Newton's equations which is induced by a plane rotation.

For the 3-body problem there are five relative equilibria: three found by Lagrange, two by Euler. For 4 bodies the finiteness is unknown.

In [Smale, 1970], I interpreted the relative equilibria as critical points of a function induced by the potential of the planar n -body problem. More precisely, the relative equilibria correspond to the critical points of

$$\hat{V}: (S_k - \Delta)/SO(2) \rightarrow \mathbb{R}$$

where $S_k = \{x \in (\mathbb{R}^2)^n \mid \sum m_i x_i = 0, \frac{1}{2} \sum m_i \|x_i\|^2 = 1\}$, $\Delta = \{x \in S_k \mid x_i = x_j \text{ some } i \neq j\}$. The rotation group $SO(2)$ acts on $S_k - \Delta$, and $\hat{V}$ is induced on the quotient from the potential function

$$V(x) = \sum_{i < j} \frac{m_i m_j}{\|x_i - x_j\|}.$$

Note that $V: S_k \rightarrow \mathbb{R}$ is invariant under the rotation group $SO(2)$ and that the quotient space $S_k/SO(2)$ is homeomorphic to complex projective space of dimension $n - 2$.

Mike Shub (1970) has shown that the set of critical points is compact, and Palmore (1976) that even for $n = 4$, $\hat{V}$ may have degenerate critical points.

Relative equilibria play an important role in celestial mechanics, for example, as in the bifurcation of the angular momentum map. Moreover there are "the Trojans" in the solar system, which correspond to the Lagrange relative equilibria.

Kuz'mina (1977) has found explicit upper bounds in the generic case.

Further background may be found in [Abraham-Marsden, 1978].

Problem 7: Distribution of Points on the 2-Sphere

Let $V_N(x) = \sum_{1 \leq i < j \leq N} \log \frac{1}{\|x_i - x_j\|}$, where $x = (x_1, \dots, x_N)$, the x_i are distinct points on the 2-sphere $S^2 \subset \mathbb{R}^3$, and $\|x_i - x_j\|$ is the distance in $\mathbb{R}^3$. Denote $\min_x V_N(x)$ by V_N .

Find $(x_1, \dots, x_N)$ such that

$$V_N(x) - V_N \leq c \log N, \quad c \text{ a universal constant.} \quad (2)$$

To "find" means to give an algorithm which on input N outputs distinct $x_1, \dots, x_N$ on the 2-sphere satisfying (2). To be precise one could take a real number algorithm in the sense of BCSS (adjoining a square root computation) with halting time polynomial in N .

This problem emerged from complexity theory, jointly with Mike Shub [Shub and Smale, 1993]. It is motivated by finding a good starting polynomial for a homotopy algorithm for realizing the Fundamental Theorem of Algebra.

An $(x_1, \dots, x_N) = x$ such that $V_N(x) = V_N$ is called an N -tuple of elliptic Fekete points (see [Tsuji, 1959]).

The function V_N as a function of N satisfies

$$V_N = -\frac{1}{4} \log \left(\frac{4}{e} \right) N^2 - \frac{1}{4} N \log N + O(N).$$

It is natural also to consider the functions

$$V_N(x, s) = \sum_{i < j} \frac{1}{\|x_i - x_j\|^s}, \quad V_{N,s} = \min_x V_N(x, s),$$

x as before and $0 < s < 2$. The original $V_N(x)$, V_N correspond in a natural way to $s = 0$, and for $s = 1$, $V_N(x, 1)$ is the Coulomb potential, and $V_N(1)$ corresponds to an equilibrium position of N electrons constrained to lie on the two-sphere.

I had asked Ed Saff for some help in dealing with the main problem above. Subsequently, he and his colleagues produced a number of fine papers dealing with the subject and its ramifications. See [Kuijlaars and Saff, 1997] and [Saff and Kuijlaars, 1997] for background and further references. In [Rakhmanov, Saff, and Zhou, 1994], one can find numerical evidence ($N = 12,000$) to support these problems.

Another way of looking at our main problem here is to optimize the function

$$W_N(x) = (\exp V_N(x))^{-1} = \prod_{i < j} \|x_i - x_j\|.$$

However, as was written in [Shub and Smale, 1993], "... this may not be so easy since there are saddle points of index N (on a great circle in S^2 , evenly spaced N points, $x_1, \dots, x_N$). Also the various symmetries that W_N possesses will confuse the picture."

Problem 8: Introduction of Dynamics into Economic Theory

The following problem is not one of pure mathematics, but lies on the interface of economics and mathematics. It has been solved only in quite limited situations.

Extend the mathematical model of general equilibrium theory to include price adjustments.

There is a (static) theory of equilibrium prices in economics starting with Walras and firmly grounded in the work of Arrow and Debreu (see [Debreu, 1959]). For the trivial case of one market this amounts to the equation "supply equals demand," and a natural dynamics is easily found [Samuelson, 1971]. For several markets, the situation is complex.

There is a function called the excess demand, $Z(p) = D(p) - S(p)$, from the space of prices to the space of commodities. Both demand D and supply S are defined by aggregation over the individual agents. Economics justifies conditions on individual behavior which lead to axioms on Z . These axioms for the excess demand map $Z: \mathbb{R}_+^\ell \rightarrow \mathbb{R}^\ell$ are:

1. $Z(\lambda p) = Z(p)$, all $p = (p_1, \dots, p_\ell)$, $p_i \geq 0$, $\lambda \in \mathbb{R}$, $\lambda > 0$.
2. $\sum_{i=1}^\ell p_i Z_i(p) = 0$, Walras's Law (the total value is zero).
3. $Z_i(p) > 0$ if $p_i = 0$ (positive demand for a free good).

By (1), (2), (3), Z may be interpreted as a vector field on the intersection of the $(\ell - 1)$ -sphere with the positive orthant, pointing inward on the boundary. The existence of an equilibrium price vector p^* follows from Hopf's theorem, so that $Z(p^*) = 0$, and "supply equals demand."

Problem 8 asks for a dynamical model, whose states are price vectors (perhaps enlarged to include other economic variables). This theory should be compatible with the existing equilibrium theory. A most desirable feature is to have the time development of prices determined by the individual actions of economic agents.

I worked on this problem for several years, feeling that it was the main problem of economic theory [Smale, 1976]. See also [Smale, 1981] for background.

Problem 9: The Linear Programming Problem

Is there a polynomial-time algorithm over the real numbers which decides the feasibility of the linear system of inequalities $Ax \geq b$?

The algorithm requested by this problem is that given by a real number machine in the sense of BCSS (see also Problem 3). The system $Ax \geq b$ has as input an $m \times n$ real matrix A and vector $b \in \mathbb{R}^m$, and the problem asks, is there some $x \in \mathbb{R}^n$ with $\sum_{j=1}^n a_{ij}x_j \geq b_i$ for all $i = 1, \dots, m$? Time is measured by the number of arithmetic operations. This problem is in BCSS.

This is a decision version of the optimization problem of linear programming: Given A, b as above and $c \in \mathbb{R}^n$, decide if

$$\max_{x \in \mathbb{R}^n} c \cdot x \quad \text{subject to } Ax \geq b$$

exists, and if so, output such an x .

The famous simplex method of Dantzig provides an algorithm for both problems (over $\mathbb{R}$), but Klee and Minty showed that it was exponentially slow in the worst case. On the other hand, Borgwardt and I, with subsequent important support from Haimovich, showed that it was polynomial time on the average. For all of these things, see [Schrijver, 1986].

In terms of the Turing model of computation, using rational numbers $\mathbb{Q}$, and cost measured by "bits," there is a parallel development. Starting with ideas of Yudin and Nemirovsky, Khachian found a polynomial time algorithm (the ellipsoid method) for the linear programming problem. Subsequently Karmarkar with his "interior point method" found a practical algorithm for this problem, which he showed ran in polynomial time in the Turing model. For all these things one can see [Grötschel, Lovász, and Schrijver, 1993] as well as [Schrijver, 1986].

Closer to the main problem above over $\mathbb{R}$ is a similar problem asking for a "strongly polynomial algorithm" over $\mathbb{Q}$ for solving these linear programming problems. This notion of algorithm demands that the number of arithmetic operations as well as the number of bit operations be polynomial in the size of the input ($m \times (n + 1)$). Partial

results are due to Megiddo and especially Tardos (see [Grötschel-Lovász-Schrijver, 1993]).

For the problem over $\mathbb{R}$ there are also references [Barvinok-Vershik, 1993] and [Traub-Woźniakowski, 1982].

Problem 10: The Closing Lemma

Let p be a non-wandering point of a diffeomorphism $S : M \rightarrow M$ of a compact manifold. Can S be arbitrarily well approximated with derivatives of order r (C^r -approximation) for each r , by $T : M \rightarrow M$ so that p is a periodic point of T ?

A non-wandering point $p \in M$ is one with the property that for each neighborhood U of p there is a $k \in \mathbb{Z}$ such that $S^k U \cap U \neq \emptyset$. Here S^k is the k^{th} iterate of S ; p is a periodic point of period m if $T^m(p) = p$.

This is the discrete form of the famous “closing lemma,” which in the C^1 -case has been solved affirmatively by Charles Pugh (1967).

There is an easy C^0 -approximation with the desired property. Peixoto observed that this argument failed for C^1 -approximations, correcting a mistake of René Thom (which René has told me was his biggest mistake).

Pugh and Robinson (1983) have proved the closing lemma with C^1 -approximations for the Hamiltonian version. Peixoto gave an affirmative answer with C^r -approximations (any r) for the circle, as well as the continuous-time version for orientable 2-dimensional manifolds. Recently the closing lemma has been given additional importance by the work of Hayashi (1997); see also [Wen-Xia, 1997].

Problem 11: Is One-Dimensional Dynamics Generally Hyperbolic?

Can a complex polynomial T be approximated by one of the same degree with the property that every critical point tends to a periodic sink under iteration?

This is unsolved even for polynomials of degree 2. Here a polynomial map $T : \mathbb{C} \rightarrow \mathbb{C}$ is considered a discrete dynamical system by iteration. So if $z \in \mathbb{C}$, its orbit in time, $z = z_0, z_1, z_2, \dots$ is defined by $z_i = T(z_{i-1})$ and i may be interpreted as time (discrete). A fixed-point w of T ($T(w) = w$) is a *sink* if the derivative $T'(w)$ of T at w has absolute value less than 1. A periodic sink of T of period p is a sink for T^p . A critical point of T is just a point where the derivative of T is zero.

While the problem is now made precise, it is useful to see it in the framework of hyperbolic dynamics from the 1960's.

A fixed-point x of a diffeomorphism $T : M \rightarrow M$ is *hyperbolic* if the derivative $DT(x)$ of T at x (as a linear automorphism of the tangent space) has no eigenvalue of absolute value 1. If x is a periodic point of period p , then x is hyperbolic if it is a hyperbolic fixed-point of T^p . The notion of hyperbolic extends naturally to Ω , the closure of the set of non-wandering points (see Problem 10).

A dynamical system $T \in \text{Diff}(M)$ is called *hyperbolic* (or

satisfies Axiom A) if the periodic points are dense in Ω and Ω is hyperbolic [Smale, 1967] or [Smale, 1980]. We assume also a no-cycle condition. The work of many people, especially Ricardo Mañé, has identified hyperbolic dynamics with a strong notion of the stability of the dynamics called structural stability. There is even the beginning of a structure theory for this class of dynamics.

While hyperbolic systems constitute a large set of dynamics, an even larger set, including applied chaotic dynamics, lies beyond. The concept of hyperbolicity extends from the invertible dynamics to the case of our problem above, polynomial maps from $\mathbb{C}$ to $\mathbb{C}$. Classical complex variable theory permits recasting the problem to an equivalent one:

Can a polynomial map $T : \mathbb{C} \rightarrow \mathbb{C}$ be approximated by one which is hyperbolic?

The theory of complex one-dimensional dynamics was begun by Fatou and Julia towards the beginning of this century. In the 1960's I asked my thesis student John Guckenheimer to look at this literature and try to solve the above problem (among other things). His thesis (see [Chern and Smale, 1970]) contains the affirmative answer, but with a gap in the proof. Now the problem stands open as the fundamental problem of one-dimensional dynamics.

John's paper is one of many on the problems of this essay with a wrong proof, starting with Poincaré.

Complex 1-dimensional dynamics has become a flourishing subject and includes important contributions of Douady and Hubbard, Sullivan, Yoccoz, McMullen, among many others. See [McMullen, 1994].

There is a parallel field of real 1-dimensional dynamics of a smooth map $T : I \rightarrow I$, $I = [0, 1]$.

Problem: *Can a smooth map $T : [0, 1] \rightarrow [0, 1]$ be approximated in C^r , all $r > 1$, by one which is hyperbolic?*

About the time of Guckenheimer's thesis, I asked Ziggy Nitecki to study this problem. My earlier negligence was compounded in not catching the mistake in Nitecki's thesis (see [Chern and Smale, 1970]), which purported to give an affirmative proof.

Subsequently, Jakobson (1971) answered the problem for C^1 -approximations, but the general case remains open. See [de Melo-van Strien, 1993] for background.

Problem 12: Centralizers of Diffeomorphisms

Can a diffeomorphism of a compact manifold M onto itself be C^r -approximated, all $r \geq 1$, by one $T : M \rightarrow M$ which commutes with only its iterates?

Thus the centralizer of T in the group of diffeomorphisms, $\text{Diff}(M)$ should be $\{T^k \mid k \in \mathbb{Z}\}$.

I had started thinking about the centralizer in [Smale, 1963], but it was after Nancy Kopell's thesis with me (see [Chern and Smale, 1970]), answering the question affirmatively in case $\dim M = 1$, that I proposed this problem

[Smale, 1967]. Today it remains unsolved even for 2-dimensional manifolds M .

One may also ask if the set of diffeomorphisms of M with trivial centralizer is dense and *open* in $\text{Diff}(M)$ with the C^r -topology.

The main work on these problems has been done by Palis and Yoccoz (1989), with almost complete answers in the case of hyperbolic dynamics (see Problem 11) for any manifold.

I wrote in [Smale, 1991], “I find this problem interesting in that it gives some focus in the dark realm, beyond hyperbolicity, where even the problems are hard to pose clearly.”

Problem 13: Hilbert’s 16th Problem

Consider the differential equation in $\mathbb{R}^2$

$$\frac{dx}{dt} = P(x,y), \quad \frac{dy}{dt} = Q(x,y), \quad (3)$$

where P and Q are polynomials. Is there a bound K on the number of limit cycles of the form $K \leq d^q$, where d is the maximum of the degrees of P and Q , and q is a universal constant?

This is a modern version of the second half of Hilbert’s 16th Problem. Except for the Riemann hypothesis, it seems to be the most elusive of Hilbert’s problems.

In fact, since a paper of Petrovskii and Landis (1957) purporting to give a positive solution, the progress seems to be backwards. Earlier, Dulac (1923) claimed that the system (3) always has a finite number of limit cycles. After a gap in Petrovskii and Landis was found (see [Petrovskii and Landis, 1959]), Ilyashenko (1985) found an error in Dulac’s paper. Moreover Shi Songling (1982) found a counterexample to the specific bounds of Petrovskii-Landis for the case $d = 2$. Subsequently, two long works have appeared, independently, giving proofs of Dulac’s assertion [Écalle, 1992] and [Ilyashenko, 1991]. These two papers have yet to be thoroughly digested by the mathematical community.

Thus one has the finiteness, but no bounds. We will consider a special class where the finiteness is simple, but the bounds remain unproved.

The following corresponds to Liénard’s equation (see, e.g., [Hirsch and Smale, 1974])

$$\frac{dx}{dt} = y - f(x), \quad \frac{dy}{dt} = -x, \quad (4)$$

where $f(x)$ is a real polynomial with leading term x^{2k+1} and satisfying $f(0) = 0$.

If $f(x) = x^2 - x$, then (4) is van der Pol’s equation with one limit cycle. More generally, it can be easily shown that all the solutions of (4) circle around the unique equilibrium at $(0,0)$ in a clockwise direction. By following these curves, one defines a “Poincaré section,” $T: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ where $\mathbb{R}^+$ is the positive y -axis. The limit cycles of (4) are precisely the fixed-points of T . In various talks I raised the question of estimating the number of these fixed-points (via some new kind of fixed-point theorem?). In response, Linz, de Melo, and Pugh (1977) found examples with k different limit cycles and

conjectured this number k for the upper bound. Still no upper bound of the form $(\deg f)^q$ has been found. Because T is analytic, (4) has a finite number of limit cycles for each f .

For more background see [Browder, 1976], [Ilyashenko and Yakovenko, 1995], [Lloyd and Lynch, 1988], and [Smale, 1991].

Problem 14: Lorenz Attractor

Is the dynamics of the ordinary differential equations of Lorenz that of the geometric Lorenz attractor of Williams, Guckenheimer, and Yorke?

The Lorenz equations are:

$$\begin{aligned} \dot{x} &= -10x + 10y \\ \dot{y} &= 28x - y - xz \\ \dot{z} &= -\frac{8}{3}z + xy. \end{aligned}$$

Lorenz (1963) analysed these equations by computer to find that most solutions tended to a certain attracting set, and in so doing, he produced an important early example of “chaos.” However, mathematical proofs were lacking. This numerical work inspired the rigorous mathematical development of a geometrically defined ordinary differential equation which seems to have the same behavior (Yorke, [Williams, 1979], [Guckenheimer and Williams, 1979]). This geometric attractor has been analysed in detail and one can fairly say that it is understood.

Problem 14 asks if the dynamics of the original equations is the same as that of the geometric model. The most complete positive answer would be to describe a homeomorphism of $\mathbb{R}^3$ to $\mathbb{R}^3$ which would take solutions of the Lorenz equations to solutions (a member of the family) of the geometric attractor—or rather, of a member of the two-parameter family of geometric attractors.

An answer to this problem would be a step in establishing foundations for the field of applied dynamical systems. Up to the present time, in the equations of engineering and physics, chaos has only been established in a weaker sense, that of proving the existence of horseshoes (e.g., Melnikov, Marsden, and Holmes; see [Guckenheimer and Holmes, 1990]).

Geometric, structurally stable, chaotic attractors in dynamics are in my paper [Smale, 1967]. But these did not arise from any physical system.

Some partial results on problem 14 are due to Rychlik and Robinson, see [Robinson, 1989].

Problem 15: Navier-Stokes Equations

Do the Navier-Stokes equations on a 3-dimensional domain Ω in $\mathbb{R}^3$ have a unique smooth solution for all time?

This is perhaps the most celebrated problem in partial differential equations. Let us be a little more precise. The Navier-Stokes equations may be written

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u - \nu \Delta u + \text{grad } p = 0, \quad \text{div } u = 0,$$

where a C^∞ -map $u : \mathbb{R}_+ \times \Omega \rightarrow \mathbb{R}^3$ and $p : \Omega \rightarrow \mathbb{R}$ are to be found satisfying these equations, u prescribed at $t = 0$ and on the boundary $\partial\Omega$. Here $\mathbb{R}_+ = [0, \infty)$, $u \cdot \nabla$ is the operator $\sum_1^3 u_i \frac{\partial}{\partial x_i}$, and ν is a positive constant. See, e.g., [Chorin-Marsden, 1993] for details.

Many mathematicians have contributed toward the understanding of this problem. An affirmative answer has been given in dimension 2 and in dimension 3 for t in a small interval $[0, T]$. See [Temam, 1979] for background.

The solution of this problem might well be a fundamental step toward the very big problem of understanding turbulence. For example, it could help realize the ideas of [Ruelle and Takens 1971], which put the notion of a chaotic attractor into a model of turbulence. See also [Chorin, Marsden, and Smale, 1977].

In [Smale, 1991] I asked if the solutions of the 2-dimensional Navier-Stokes equation with a forcing term on a torus must converge to an equilibrium as time tends to infinity. Babik and Vishik (1983) had given some evidence to the contrary. Subsequently Liu (1992) provided examples to show convergence to a more complicated attractor.

Problem 16: The Jacobian Conjecture

Suppose $f : \mathbb{C}^n \rightarrow \mathbb{C}^n$ is a polynomial map with the property that the derivative at each point is non-singular. Then must f be one-to-one?

Here $f(z) = (f_1(z), \dots, f_n(z))$, and each f_i is a polynomial in n variables. The derivative of f at z , $Df(z) : \mathbb{C}^n \rightarrow \mathbb{C}^n$, may be thought of as the matrix of partial derivatives, and the non-singularity condition as $\text{Det } Df(z) \neq 0$.

If f is indeed injective, then it is surjective and has an inverse which is a polynomial map. For an elementary proof of this see [Rudin, 1995].

The problem goes back to the 1930's. See the excellent surveys [Bass, Connell, and Wright, 1982] and [van den Essen, 1997] for the importance, background, and related results.

Problem 17: Solving Polynomial Equations

Can a zero of n complex polynomial equations in n unknowns be found approximately, on the average, in polynomial time with a uniform algorithm?

The final theorem in my five-paper series with Mike Shub [Shub and Smale, 1994] is exactly this result without the word "uniform."

I review the definitions. Consider $f : \mathbb{C}^n \rightarrow \mathbb{C}^n$, $f(z) = (f_1(z), \dots, f_n(z))$, $z = (z_1, \dots, z_n)$ where each f_i is a polynomial in n variables of degree d_i . It is reasonable to make the f_i into homogeneous polynomials by adding a new variable z_0 , work in the corresponding projective space, and then translate the algorithm and results back to the initial affine problem.

"Approximately" can be defined intrinsically using Newton's method, and is necessary because of Abel, Galois, *et al.* Time is measured by the number of arithmetic operations and comparisons, $\leq$, using real machines (as in Problem 3), if one wants to be formal.

A probability measure must be put on the space of all such f , for each $d = (d_1, \dots, d_n)$, and the time of an algorithm is averaged over the space of f . Is there such an algorithm where the average time is bounded by a polynomial in the number of coefficients of f (the input size)?

In [Shub and Smale, 1994] it is proved that this can be done, but the algorithm is different for each d and even for each desired probability of success. A uniform algorithm is one that is independent of d (d is part of the input).

Certainly finding zeros of polynomials and polynomial systems is one of the oldest and most central problems of mathematics. This problem asks if under some conditions, specified in the problem, it can be solved systematically by computers. If there is no polynomial-time way of doing it, then no computer will ever succeed.

There is a more recent development putting the problem of zeros of polynomials into a universal role. The Hilbert Nullstellensatz (as a decision problem) is NP-complete over any field (see Problem 3).

A similar, more difficult, problem may be raised for the real numbers.

Problem 18: Limits of Intelligence

What are the limits of intelligence, both artificial and human?

Penrose (1991) attempts to show some limitations of artificial intelligence. His argumentation brings in the interesting question whether the Mandelbrot set is decidable (dealt with in [Blum and Smale, 1993]) and implications of the Gödel incompleteness theorem.

However, a broader study is called for, one which involves deeper models of the brain, and of the computer, in a search of what artificial and human intelligence have in common, and how they differ. I would look in a direction where learning, problem-solving, and game theory play a substantial role, together with the mathematics of real numbers, approximations, probability, and geometry.

I hope to expand on these thoughts on another occasion.

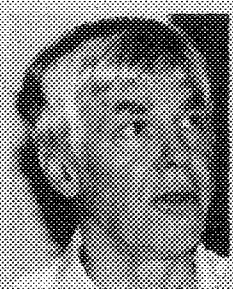
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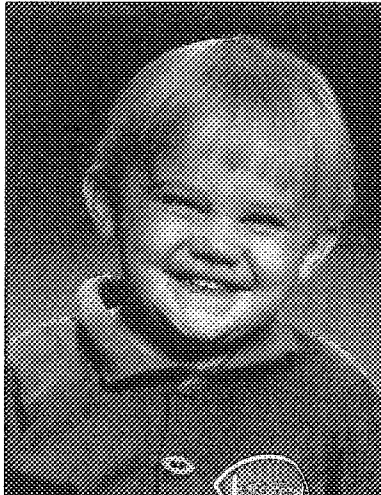
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